



**JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR**

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A place to learn; A chance to grow"

2025-2026 (Odd Semester)

MSE-2 EXAMINATION

Course: B. Tech. in CSE

Subject: Interactive Graphics

Max Marks: 20

Year/Semester: 5th Semester (3rd Year)

Subject Code: CS5E001C

Duration: - 1 Hr.

Date: 03/10/2025

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

- CO 1: Understand the types of display systems used in computer graphics.
CO 2: Apply graphical concepts like screen/user coordinates and data structures.
CO 3: Analyze different display technologies and evaluate their advantages.
CO 4: Explain digital frame buffers and display processors in context.
CO 5: Use knowledge of graphical data structures in application-based problems.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	What is a light pen?	CO 3	1
Q.2	What is homogeneous coordinate?	CO 4	1
Q.3	Define rubber-band line technique.	CO 3	1
Q.4	What is line clipping?	CO 4	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence).

Q.5	Differentiate between mouse and trackball	CO 3	2
Q.6	Explain point clipping with example.	CO 4	2
Q.7	What is window-to-viewport transformation?	CO 4	2
Q.8	Write short notes on zooming and panning.	CO 3	2

SECTION C

1. Solve any two questions.

Q.9	Illustrate Cohen-Sutherland line clipping algorithm with an example.	CO 4	4
Q.10	Explain composite transformations with neat matrix representation	CO 5	4
Q.11	Discuss various interactive positioning techniques (elastic lines, linking, windowing, scissoring).	CO 3	4